NAIVE DERIVED COMPLETION AND DERIVED COMPLETION AGREE EXACTLY WHEN THE F-POWER TORSION IS BOUNDED

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Editorial note. The counted statement span displays the derived completion via the two-term system (A -> f^n -> A); that displayed material is delivered verbatim. The five references inside the counted statement and proof spans are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The lifting statement identifying the pro-object of multiplication maps with the pro-object of quotient rings when the torsion is bounded. invoked at proof lines 26825-26827. Gives the forward implication.
- The exactness statement for tensor products with R lim and the vanishing criterion. invoked at proof lines 26829-26834. Turns the isomorphism condition into a vanishing condition for all K.
- Vanishing of the derived limit of direct sums in the relevant range. invoked at proof lines 26836-26838. Forces the system of f-power torsion modules to be pro-zero.
- The derived completion defined via the Koszul-type system (A -> f^n -> A). referred to at statement lines 26816-26819. Fixes the left-hand functor of the natural transformation being compared.

Source reference. Original lemma 23405: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23405–23405. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 23527: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23527–23527. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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Source reference. Original lemma 26388: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26388–26388. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 26782: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26782–26782. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 23915: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23915–23915. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring and $f \in A$. Set $I = (f)$. In this situation we have the naive derived completion $K \mapsto K' = R\lim(K \otimes^{\mathbf{L}}_A A/f^nA)$ and the derived completion*

$$K \mapsto K^\wedge = R\lim(K \otimes^{\mathbf{L}}_A (A \xrightarrow{f^n} A))$$

of Lemma 26388. The natural transformation of functors $K^\wedge \to K'$ is an isomorphism if and only if the f-power torsion of A is bounded.

Proof. If the f-power torsion is bounded, then the pro-objects $\{(f^n : A \to A)\}$ and $\{A/f^nA\}$ are isomorphic by Lemma 26782. Hence the functors are isomorphic by Lemma 23405. Conversely, we see from Lemma 23915 that the condition is exactly that

$$R\lim(K \otimes^{\mathbf{L}}_A A[f^n])$$

is zero for all $K \in D(A)$. Here the maps of the system $(A[f^n])$ are given by multiplication by f. Taking $K = A$ and $K = \bigoplus_{i \in \mathbf{N}} A$ we see from Lemma 23527 this implies $(A[f^n])$ is zero as a pro-object, i.e., $f^{n-1}A[f^n] = 0$ for some n, i.e., $A[f^{n-1}] = A[f^n]$, i.e., the f-power torsion is bounded. □